

# Abhay Murthy

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## EDUCATION

### University of California, Berkeley (M.E.T Program)

Berkeley, CA

*B.S. Bioengineering & B.S. Business Administration*

*Aug. 2026 – May 2030*

### Liberty High School

Frisco, TX

*Rank: 4/444 | GPA: 5.559 | SAT: 1570*

*Sep. 2022 – May 2026*

- **Awards:** Jane Street M3C Intl Finalist & Top Presenter, DECA Intl (ICDC) Winner, BPA National Winner, Invited Presenter @ SfN
- **Coursework:** Multivariable Calculus, Differential Equations, Data Structures & Algorithms, AP Calculus BC, AP Statistics, AP Physics C, AP Biology, AP Computer Science A, AP Macroeconomics, Statistical Learning with Python (Stanford)

## EXPERIENCE

### Boston University

Jun. – Aug. 2025

*Research Intern (RISE)*

*Boston, MA*

- Developed an ODE-based computational model of neuron–neurotrophin interactions integrating receptor affinities, cleavage enzymes, and neurodegeneration dynamics; applied MM-MD on an HPC cluster to estimate binding rates for canonical and non-canonical BDNF receptor complexes
- Proposed a novel biomarker framework identifying “tipping point” proBDNF:mBDNF ratios for early Alzheimer’s detection; accepted and presented at Society for Neuroscience (SfN) Annual Meeting 2025
- Co-authored literature review on the blood-brain barrier, BDNF signaling, and drug design for BBB penetration

### University of Texas at Dallas

Jun. – Aug. 2024

*Data Analytics Research Intern*

*Richardson, TX*

- Engineered a Python script to scrape and clean ~2,000 data points from Transfermarkt.us across 30 years; analyzed longitudinal trends (1990–2019) with Tableau, investigating multivariate correlations between age, position, league, and individual accolades
- Tested five hypotheses on key predictive traits of Ballon d’Or winners, identifying 4 statistically significant factors

## PROJECTS

### Neural Network From Scratch | *Python, NumPy, Linear Algebra, Calculus*

Present

- Built a fully custom neural network from basic neuron operations to multi-layer architectures without ML libraries
- Implemented ReLU, Softmax, Sigmoid, and Linear activations with vectorized forward passes
- Derived and coded backpropagation manually using multivariable calculus (chain rule) and matrix operations

### Wake Word Detection Model | *Python, TensorFlow, Librosa, Pandas*

Aug. 2024

- Built a custom audio data collection and labeling pipeline; developed preprocessing pipeline using Librosa to extract MFCC features from audio files
- Trained and optimized a neural network using TensorFlow Keras, achieving 100% detection accuracy

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, JavaScript, HTML, CSS, MATLAB, GROMACS

**Frameworks & Libraries:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Librosa, React, Next.js

**Developer Tools:** Git, GitHub, Tableau, ANSYS, Microsoft 365

**Domain Skills:** Computational Neuroscience, Neural Systems, Machine Learning, Data Mining, Data Visualization, Databases, Data Structures & Algorithms